THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 41	LCM plus the common remainder
2	**(d)** 57	71 multiples up to 500 minus 14 multiples below 100
3	**(b)** 112	Count multiples from 100 to 999
4	**(c)** 111	Test palindromes 101, 111, 121, ...
5	**(d)** 1	Divisible by 10 and 9
6	**(a)** 5	Apply the routine and count
7	**(d)** 99	9 and 11 have no common factor
8	**(b)** 67	LCM plus the common remainder
9	**(a)** Both A and R are true and R is the correct explanation of A.	$4+2+3+6=15$, which is divisible by 3; R is the test used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	The last two digits 24 are divisible by 4, so A is true. R is also a true test but is about 8, so it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	LCM(20, 25) $=100$ (1); number $=100+3=103$ (1).	1+1
12	72 multiples up to 800 (1); minus 13 below 150, giving 59 (1).	1+1
13	Middle number $=45\div5=9$ (1); numbers 7, 8, 9, 10, 11 (1).	1+1
14	LCM(24, 36) $=72$ (1); number $=72+5=77$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$247247=247\times1001$ (1); $1001=7\times11\times13$ (1); so divisible by 7, 11 and 13 (quotients 35321, 22477, 19019) (1).	1+1+1
16	LCM $=24$ (1); 9999 divided by it leaves remainder 15 (1); number 9984 (1).	1+1+1
17	LCM $=24$ (1); multiples from 100 to 999: $41-4$ (1); 37 (1).	1+1+1
	OR: $358358=358\times1001$ (1); $1001=7\times11\times13$ (1); so divisible by 7, 11 and 13 (quotients 51194, 32578, 27566) (1).	1+1+1
18	LCM $=60$ (1); 9999 divided by it leaves remainder 39 (1); number 9960 (1).	1+1+1
	OR: LCM $=36$ (1); multiples from 100 to 999: $27-2$ (1); 25 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Digit sum 18, divisible by 9, so yes (1). (ii) Last two digits 36 are divisible by 4, so yes (1). (iii) Odd places $6+2=8$, even places $3+7=10$; difference 2, so not divisible by 11 (1). (iv) $5+2+d+4=11+d$ must be 18, so $d=7$ (1); 5274 is even and its digit sum 18 is divisible by 3, so it is divisible by 6 ($5274=6\times879$) (1).	1+1+1+2
20	Splitting 90 into coprime factors and applying the rules (2); possibilities: a = 1, b = 0 (number 6120) (2); verification (1).	2+2+1
	OR: 641-146=495 (1); reverse of 495 is 594 (1); 495+594=1089 (1); in general the difference is $99(a-c)$, whose digits are $(a-c-1), 9, (10-a+c)$ and the sum is always 1089 (2).	1+1+1+2

Mistakes to avoid:

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 212	Test palindromes 101, 111, 121, ...
2	**(b)** 4	Divisible by 10 and 9
3	**(a)** 5	Apply the routine and count
4	**(a)** 72	8 and 9 have no common factor
5	**(c)** 46	LCM plus the common remainder
6	**(d)** 63	66 multiples up to 1000 minus 3 multiples below 50
7	**(b)** 53	Count multiples from 100 to 999
8	**(b)** 222	Test palindromes 101, 111, 121, ...
9	**(c)** A is true but R is false.	A is true. R is false: the test for 11 uses the difference of the sums of digits at odd and even places (e.g. 121 has digit sum 4 but is divisible by 11).
10	**(d)** A is false but R is true.	$2+3+4+5=14$ is not divisible by 9, so A is false; R is a true test.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	107 multiples up to 1500 (1); minus 17 below 250, giving 90 (1).	1+1
12	Middle number $= 80 \div 5 = 16$ (1); numbers 14, 15, 16, 17, 18 (1).	1+1
13	LCM(28, 42) $= 84$ (1); number $= 84 + 9 = 93$ (1).	1+1
14	100 multiples up to 600 (1); minus 12 below 75, giving 88 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$429429 = 429 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 61347, 39039, 33033) (1).	1+1+1
16	LCM $= 90$ (1); 9999 divided by it leaves remainder 9 (1); number 9990 (1).	1+1+1
17	LCM $= 70$ (1); multiples from 100 to 999: $14 - 1$ (1); 13 (1).	1+1+1
	OR: $516516 = 516 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 73788, 46956, 39732) (1).	1+1+1
18	LCM $= 84$ (1); 9999 divided by it leaves remainder 3 (1); number 9996 (1).	1+1+1
	OR: LCM $= 60$ (1); multiples from 100 to 999: $16 - 1$ (1); 15 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $532 - 235 = 297$ (1). (ii) $972 - 279 = 693$ (1). (iii) $963 - 369 = 594$, $954 - 459 = 495$, so 495 (1). (iv) $4321 - 1234 = 3087$, $8730 - 0378 = 8352$, $8532 - 2358 = 6174$ (1); the number reached is 6174 in 3 steps (1).	1+1+1+2
20	Splitting 45 into coprime factors and applying the rules (2); possibilities: a = 0, b = 0 (number 8010), a = 4, b = 5 (number 8415), a = 9, b = 0 (number 8910) (2); verification (1).	2+2+1
	OR: 723-327=396 (1); reverse of 396 is 693 (1); 396+693=1089 (1); in general the difference is $99(a-c)$, whose digits are $(a-c-1), 9, (10-a+c)$ and the sum is always 1089 (2).	1+1+1+2

Mistakes to avoid:

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 35	5 and 7 have no common factor
2	**(a)** 57	LCM plus the common remainder
3	**(c)** 69	80 multiples up to 2000 minus 11 multiples below 300
4	**(b)** 41	Count multiples from 100 to 999
5	**(c)** 161	Test palindromes 101, 111, 121, ...
6	**(c)** 8	Divisible by 10 and 9
7	**(c)** 7	Apply the routine and count
8	**(a)** 56	7 and 8 have no common factor
9	**(a)** Both A and R are true and R is the correct explanation of A.	From the right: odd places $8+7=15$, even places $2+2=4$; difference 11, so divisible ($2728 = 11 \times 248$).
10	**(c)** A is true but R is false.	$5432-2345=3087$, $8730-0378=8352$, $8532-2358=6174$, so A is true. R is false: 495 is reached for three-digit numbers, while four-digit numbers reach 6174.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Middle number $= 115 \div 5 = 23$ (1); numbers 21, 22, 23, 24, 25 (1).	1+1
12	LCM(30, 45) $= 90$ (1); number $= 90 + 8 = 98$ (1).	1+1
13	40 multiples up to 5000 (1); minus 7 below 1000, giving 33 (1).	1+1
14	Middle number $= 150 \div 5 = 30$ (1); numbers 28, 29, 30, 31, 32 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$683683 = 683 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 97669, 62153, 52591) (1).	1+1+1
16	LCM $= 240$ (1); 9999 divided by it leaves remainder 159 (1); number 9840 (1).	1+1+1
17	LCM $= 84$ (1); multiples from 100 to 999: $11 - 1$ (1); 10 (1).	1+1+1
	OR: $791791 = 791 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 113113, 71981, 60907) (1).	1+1+1
18	LCM $= 108$ (1); 9999 divided by it leaves remainder 63 (1); number 9936 (1).	1+1+1
	OR: LCM $= 144$ (1); multiples from 100 to 999: $6 - 0$ (1); 6 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Yes (1). (ii) $57+75=132$ (1). (iii) $132+231=363$, a palindrome (1). (iv) $abba = 1000a + 100b + 10b + a = 1001a + 110b = 11(91a + 10b)$ (1); $3663 = 11 \times 333$ (1).	1+1+1+2
20	Splitting 18 into coprime factors and applying the rules (2); possibilities: a = 0, b = 2 (number 2052), a = 2, b = 0 (number 2250), a = 3, b = 8 (number 2358), a = 5, b = 6 (number 2556), a = 7, b = 4 (number 2754), a = 9, b = 2 (number 2952) (2); verification (1).	2+2+1
	OR: 971-179=792 (1); reverse of 792 is 297 (1); 792+297=1089 (1); in general the difference is $99(a-c)$, whose digits are $(a-c-1), 9, (10-a+c)$ and the sum is always 1089 (2).	1+1+1+2

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